

a Form a causal VICReg target

Temporal invariance
$$r_t^{\text{time}} = I_t(z_t - p_t)$$

Variance
$$r_t^{\text{var}} = -\frac{\lambda_V}{\lambda_T} g_t \odot (z_t - \mu_t)$$

Covariance
$$r_t^{\text{cov}} = \frac{\lambda_C}{2\lambda_T} \text{offdiag}(M_t)(z_t - \mu_t)$$

$$r_t = r_t^{\text{time}} + r_t^{\text{var}} + r_t^{\text{cov}}$$
$$\hat{z}_t = z_t - r_t$$

b Define the neuron state

Exact preactivation residual

$$b_t = (z_t - \hat{z}_t) \odot \mathbf{1}[a_t > 0]$$

one correction

$$s_t = b_t - \eta \kappa M_t b_t$$

$s_{t,j}$ is the postsynaptic value
used by neuron j for this update.

c Update local synapses

Feedforward
$$W^+ = (1 - \alpha \lambda_W) W - \alpha c s_t x_t^\top$$

Lateral
$$M^+ = \rho M + (1 - \rho) Q_t$$
$$Q_t = (1 - \gamma) \text{offdiag}(\bar{z}_t \bar{z}_t^\top) + \gamma s_t s_t^\top$$

Feedforward endpoints: x_i, s_j
Lateral endpoints: $\bar{z}_j, \bar{z}_k$ and s_j, s_k

One observation at a time

detached state

no error signal from a later layer